Assignment-9

Q.2.



$$P_{av} = \frac{1}{5} \times P_{max}$$

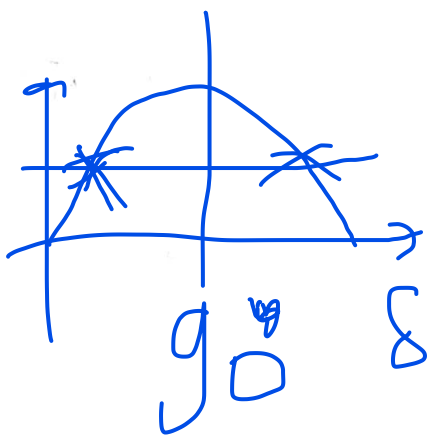
$$\frac{(1-\delta^*)\delta^* V_0 V_{in}}{2 \times X_{fm}} = \frac{1}{5} \times \frac{V_0 V_{in}}{2 \times X_{fm}}$$

$$\Rightarrow \delta^* - \delta^{*2} = \frac{1}{20}$$

$$\Rightarrow \delta^{*2} - \delta^* + \frac{1}{20} = 0$$

$$\Rightarrow \delta^* = \frac{+1 \pm \sqrt{1 - 4/20}}{2}$$

$$= \frac{1 \pm \sqrt{4/5}}{2}$$



$$\delta^* = 0.944, \ 0.0527$$

$$\Rightarrow \delta = 0.944 \times 180, \ 0.0527 \times 180$$

$$\delta = 170.46°, \ 9.486°$$

$$\Rightarrow \boxed{\delta = 9.486}$$

Q.3.

$$\Delta B = \frac{d_1}{2n_1 A_c} \Rightarrow A_c = \frac{400}{0.1 \times 2 \times 10} = 200 \ m^2$$

Q.4.

$$P_w = a\left(\frac{1}{\Delta B}\right)^2 \quad ; \quad P_{core} = b(\Delta B)^2 \quad (\alpha, \beta = 2)$$

$\Rightarrow$ at ΔB optimum value

$$\frac{dP_w}{d(\Delta B)} = -\frac{dP_{core}}{d(\Delta B)}$$

$$\Rightarrow \quad a \cdot \Delta B^{-3} \cdot 1\cancel{x} = \cancel{+}\, b \times \cancel{2} \times \Delta B$$

$$\Rightarrow \quad \frac{a}{b} = \Delta B^4$$

$$\Rightarrow \quad \Delta B = \left(\frac{a}{b}\right)^{1/4}$$

$$\Rightarrow \quad \Delta B = (a/b)^{0.25}$$

$\hookrightarrow$ (B) option

Q.7.

given $\delta = 60°$

$\Rightarrow \delta^* = \frac{60}{180} = 1/3$

$$P_{av} = \frac{\delta^*(1-\delta^*) V_0 V_{in}}{2n L f_{sw}}$$

$$= \frac{1/3 \times 2/3 \times 800 \times 200}{2 \times 4/2 \times 10 \times 10^{-6} \times 100 \times 10^3}$$

$$\boxed{P_{av} = 8.88 \text{ kW}}$$

Q.8.

$$P_{max} = \frac{V_0 V_{in}}{8 n L f_{sw}} = \frac{400 \times 400}{8 \times 4/1 \times 100 \times 10^{-6} \times 100 \times 10^3}$$

$$\boxed{P_{max} = 0.5 \text{ kW}}$$

Q.9. flat top means, $V_{in} = V_0/n$

$\Rightarrow V_0 = \frac{N_s}{N_p} V_{in}$

$\Rightarrow \boxed{V_0 = V_{in}/3}$ $\longrightarrow$ (B) option